Solution Requirements

AI Learning Assistant

Project Title

AI Learning Assistant

Domain

Artificial Intelligence | Education Technology

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1. Introduction

The Solution Requirements document defines the functional and non-functional requirements necessary for developing the AI Learning Assistant. These requirements ensure that the system meets user expectations, performs efficiently, and provides a reliable learning experience.

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2. Functional Requirements

Functional requirements describe what the system should do.

FR-1: User Input

The system shall allow users to enter study-related questions, thoughts, or messages through a text interface.

FR-2: Emotion Detection

The system shall analyze the user's text and identify the emotional state (such as Happy, Sad, Angry, Stressed, Anxious, Motivated, Confused, or Neutral).

FR-3: AI Response Generation

The system shall send the detected emotion and user input to Google's Gemini AI and generate personalized study guidance.

FR-4: Response Display

The system shall display the AI-generated response to the user in real time.

FR-5: Interaction Logging

The system shall store the timestamp, user input, detected emotion, confidence score, and AI response in a CSV file.

FR-6: Dashboard Analytics

The system shall display analytics such as emotion distribution and interaction history using charts and tables.

FR-7: Error Handling

The system shall display appropriate error messages if invalid input is provided or the AI service is unavailable.

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3. Non-Functional Requirements

Performance

- The system should generate responses within a few seconds.
- Emotion detection should be efficient and accurate.

Reliability

- The application should operate consistently without unexpected failures.

Usability

- The interface should be simple, clean, and easy to use.

Scalability

- The system should support future enhancements such as voice emotion detection and facial emotion recognition.

Security

- The Gemini API key should be stored securely using a `.env` file.
- User interaction data should be protected from unauthorized access.

Maintainability

- The code should be modular and easy to maintain.

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4. Hardware Requirements

Component	Requirement
Processor	Intel Core i3 or above
RAM	Minimum 4 GB (8 GB Recommended)
Storage	500 MB Free Space
Internet	Required for Gemini AI API
Display	1366 × 768 Resolution or Higher

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5. Software Requirements

Software	Purpose
Windows 10/11	Operating System

Python 3.10 or above	Programming Language
Visual Studio Code	Code Editor
Streamlit	User Interface
Google Gemini API	AI Response Generation
Pandas	Data Processing
Matplotlib	Data Visualization
Python-dotenv	Environment Variable Management
Git & GitHub	Version Control

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6. System Constraints

- Internet connectivity is required for accessing the Gemini API.
- Emotion detection accuracy depends on the user's text input.
- The current version supports only text-based emotion detection.

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7. Assumptions

- Users provide meaningful text input.
- The Gemini API is available during system usage.
- The application runs on a compatible operating system with Python installed.

8. Benefits of the Proposed Solution

- Personalized learning support.
- Emotion-aware AI interaction.
- Improved student motivation.
- Better learning experience.
- Easy-to-use interface.
- Scalable architecture for future enhancements.

9. Conclusion

The Solution Requirements document outlines the essential functional and non-functional requirements needed to develop the AI Learning Assistant. These requirements ensure that the system is reliable, user-friendly, secure, and capable of providing personalized learning assistance based on students' emotional states.